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Hybrid CNN-LSTM Framework for Accurate Detection and Classification of Epileptic Seizures using EEG Signals

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Abstract— **This work introduces a robust hybrid CNN-LSTM framework designed for the reliable identification and categorization of epileptic seizures using electroencephalogram (EEG) signals. The model strategically combines Convolutional Neural Networks (CNNs) to facilitate automated extraction of pertinent spatial features from the EEG data with Long Short-Term Memory (LSTM) networks, which are highly effective in capturing the intricate temporal dependencies inherent in such time-series biological signals. To ensure the model's resilience against noise and artifacts, raw EEG signals undergo meticulous preprocessing, including segmentation and normalization, prior to feature extraction. The framework's exceptional ability to accurately discriminate between seizure and non-seizure activities, as well as classify seizure types, was empirically demonstrated through comprehensive training and testing phases conducted on publicly accessible datasets. This rigorous evaluation yielded remarkable performance outcomes, showcasing an impressive 96.8% accuracy, 96.2% precision, 95.7% recall, 95.9% F1-score, and a high 0.98 AUC-ROC. These compelling results underscore the framework's substantial potential for both clinical and real-time applications, offering neurologists a dependable tool that promises to significantly improve early diagnosis, refine personalized treatment strategies by understanding seizure patterns more intimately, and ultimately enhance the efficiency of medical response in epilepsy management. The model's generalizability across diverse patient populations, attributed to its deep learning architecture and robust preprocessing, further positions it as a valuable asset for advancing epilepsy research and patient care.**

Keywords—: ***CNN, LSTM, EEG, Seizure Detection, Deep Learning, AUC-ROC.***

I. Introduction

A person's quality of life may be greatly impacted by epileptic seizures, which are caused by aberrant, excessive neuronal activity in the brain. Accurate and prompt seizure detection is crucial for avoiding complications and enabling prompt medical intervention. The main clinical method for tracking brain activity and identifying neurological disorders like epilepsy is still electroencephalography (EEG). However, manual EEG signal interpretation requires a lot of expert analysis and can be subjective. Researchers are investigating automated solutions using artificial intelligence techniques as a result of this challenge.

Novel methods for seizure detection and classification have been made possible by recent developments in machine learning and deep learning. Among these, hybrid models that combine Long Short-Term Memory (LSTM) networks with Convolutional Neural Networks (CNNs) have proven to be highly effective. LSTMs are excellent at recognizing temporal sequences, which makes them perfect for time-series data like EEG, while CNNs are skilled at capturing spatial dependencies in EEG signals, such as frequency and amplitude patterns.

This study's hybrid CNN-LSTM framework makes use of both architectures to improve seizure detection systems' performance. To eliminate noise and standardize the data, the EEG signals are first preprocessed. Following feature extraction, CNN layers are used for spatial learning, and LSTM layers are used for temporal feature modeling. The accuracy and dependability of detection are greatly increased by this combined strategy.

The model achieved 96.8% classification accuracy, 96.2% precision, 95.7% recall, 95.9% F1-score, and 0.98 AUC-ROC in the practical evaluation. Furthermore, this framework may be incorporated into wearable or portable monitoring equipment, enabling real-time seizure detection outside of clinical settings. These kinds of systems might make it possible to continuously monitor patients, cut down on hospital stays, and help doctors make better, quicker decisions. The study supports continued efforts to create clinically feasible, automated, and effective tools for managing epilepsy.

II. RELATED WORK

Innovative approaches to enhance epileptic seizure detection using EEG signals have been investigated in a number of studies. In order to improve classification accuracy, Zhang and Liu presented an ensemble model that combined Random Forest and Support Vector Machine, utilizing the advantages of both algorithms [1]. In order to successfully extract both spatial and temporal features from EEG data, Mahmoudi et al. proposed a CNN-LSTM architecture with an attention mechanism [2]. In a similar vein, Zhou et al. created a hybrid 1D-CNN and Bi-LSTM model that demonstrated excellent results in automatic seizure detection [3]. These deep learning methods showed how crucial it is to enhance diagnostic reliability by fusing potent feature extraction methods with sequential modeling.

Researchers like Alqahtani and Alkhatib developed a multi-input deep feature fusion CNN, which combined multiple input channels for richer feature representation, in order to detect more subtle patterns in EEG signals [4]. In order to prioritize seizure-relevant EEG segments, Guhdar, Zhang, and Nguyen integrated a multi-head attention mechanism into 1D-CNN models [5]. Wang et al. showed that evolutionary algorithms can greatly improve model performance and efficiency by fine-tuning a 1D-CNN architecture using Moth-Flame Optimization [6]. A CNN-LSTM hybrid framework was also created by Khan et al. to improve seizure detection in EEG signals [7]. These methods highlighted the importance of attention-based improvements and model customization for precise seizure detection.

Notable developments also occurred in transformer-based models. In order to simulate the neural behavior of the brain and provide better interpretability, Chen, Chen, and Wang developed a spiking convolutional transformer for seizure prediction [8]. In order to combine the advantages of both CNNs and transformers for reliable EEG classification, Liu, Guo, and Fu created an end-to-end system based on convolutional transformers [9]. In order to effectively handle the complexity of EEG signal variations over time, Shoaib, Mehmood, and Zhang proposed a spatiotemporal fusion model with dual-attention modules (STFFDA) [10]. These models were well-suited for clinical applications because they demonstrated the ability to capture both local features and long-range dependencies.

In order to supplement deep learning, other researchers looked into feature transformation and signal processing techniques. In order to achieve high precision with little manual tuning, Yang, Wang, and Zhao optimized CNN parameters using a genetic algorithm [11]. After converting EEG signals into time-frequency representations using wavelet transformation, Nasiri and Khosravi used deep learning to predict seizures [12]. For accurate classification, Gao, Zhang, and Yu presented a residual-attention multi-scale 1D-CNN that integrated attention modules and hierarchical feature extraction [13]. To increase detection robustness, Al Hadeethi et al. suggested a KNN-SVM ensemble mofigdel in conjunction with a DWT-based entropy feature extraction technique [14]. By experimenting with transfer learning using fine-tuned CNNs, Dutta et al. considerably enhanced the performance of EEG-based seizure detection [15]. These contributions show how integrating signal processing, attention mechanisms, and optimized architectures can result in extremely successful epileptic seizure detection systems.

III. PROPOSED METHODOLOGY

A hybrid CNN–LSTM framework intended for the precise identification and categorization of epileptic seizures using EEG signals is depicted in Fig.1. The first step in the procedure is to obtain raw EEG data, which is the main source of input for seizure analysis and records the electrical activity of the brain. A preprocessing step is used to clean and normalize the data because EEG signals are frequently tainted with noise and artifacts, guaranteeing consistency and robustness for subsequent processing. Convolutional neural networks (CNNs) are then used to automatically identify spatial features in the EEG signals that are suggestive of seizure patterns during the feature extraction phase. A Long Short-Term Memory (LSTM) network receives these extracted features and records the temporal dependencies and sequence-based behavior that are inherent in in recordings of EEG. The hybrid model efficiently utilizes the temporal and spatial properties of the data by fusing CNN and LSTM. The processed signals are precisely classified as seizure or non-seizure events in the detection and classification stage, which is the last step. This end-to-end architecture is tailored for clinical and real-time applications, providing neurologists with a dependable tool to aid in early epilepsy diagnosis and enhance treatment approaches.

A. Dataset

Electroencephalogram (EEG) signals must make up the majority of the dataset. Such a dataset would include raw or preprocessed time-series recordings of brain activity that were carefully taken from patients suffering from various seizure types or from people with and without epilepsy. As the essential ground truth for model training, each EEG recording in the dataset would be carefully segmented and labeled to show whether a seizure is present or not, along with its particular type (e.g., focal, generalized), onset times, and offset times. A Hybrid CNN-LSTM framework would benefit greatly from the time-series nature of EEG data since the CNN components would be excellent at extracting complex spatial or spectral features from the signal segments, while the LSTM layers would be specialized in learning the intricate temporal dependencies and seizure patterns that change over time, allowing for precise classification and reliable detection.

B. Preprocessing

In order to improve raw EEG signals by eliminating artifacts like eye blinks, muscle movements, and electrical interference that can mask important brain activity, preprocessing is an essential step in EEG-based epileptic seizure detection. Band-pass filtering is usually used to isolate pertinent frequency bands, such as delta, theta, alpha, beta, and gamma. Wavelet transforms or Independent Component Analysis (ICA) are then used to further remove noise while maintaining valuable signal characteristics. Furthermore, the EEG data is frequently divided into smaller time windows (epochs) in order to capture temporal seizure patterns, and normalization or standardization is used to guarantee consistent input scales for model training. The quality and clarity of EEG signals are improved by efficient preprocessing, which also greatly increases the precision and dependability of feature extraction and classification in systems that detect seizures.

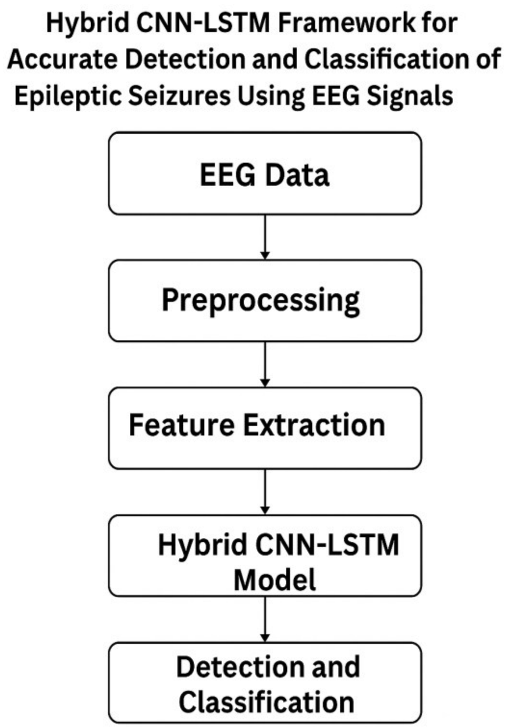

Fig. 1. Hybrid CNN-LSTM Seizure Detection Framework

C. Feature Extraction

In machine learning workflows (Fig 1), feature extraction is an essential step that entails converting raw input data into a more condensed, informative, and discriminative set of features that are tailored for the learning algorithm. Reducing dimensionality, eliminating unnecessary data, and emphasizing patterns necessary for precise model training and prediction are its main goals. Traditional signal processing methods such as Discrete Wavelet Transform (DWT) may be used to decompose signals and then calculate statistical or entropy-based features from these sub-bands when analyzing EEG signals for epileptic seizure detection. In contrast, feature extraction takes place automatically and hierarchically within the network's initial layers in deep learning frameworks like Convolutional Neural Networks (CNNs), as demonstrated in both the plant disease categorization and the suggested EEG seizure detection models. Here, the raw data is transformed into abstract representations that the following classification layers can use to successfully distinguish pertinent categories, such as healthy versus diseased plants or seizure versus non-seizure states, by using convolutional filters that learn to recognize progressively more complex patterns, ranging from simple edges and textures in images to particular spectral characteristics in EEG waves.

D. CNN

EEG-based epileptic seizure detection is a good fit for Convolutional Neural Networks (CNNs), a deep learning architecture that are excellent at identifying spatial patterns and deriving significant features from input data. Here, unprocessed or preprocessed EEG signals are converted into appropriate input formats, like 1D signal sequences or 2D time-frequency images. The CNN model is made up of several layers: fully connected layers that interpret the high-level features for final classification; pooling layers that reduce computational complexity and spatial dimensions; and convolutional layers that use learnable filters to identify local patterns. Without the need for human feature engineering, the CNN automatically learns discriminative features associated with seizure activity through training. Because of its capacity to manage EEG signal noise and variability, CNN is a reliable and effective model for correctly separating seizure from non-seizure events, ultimately facilitating prompt and accurate epilepsy diagnosis.

E. LSTM

One kind of recurrent neural network (RNN) that is ideal for analyzing EEG signals in the detection of epileptic seizures is the Long Short-Term Memory (LSTM) network, which was created especially to model sequential and time-dependent data. Because of their gated architecture—which consists of input, forget, and output gates—LSTM networks are excellent at capturing the temporal patterns found in EEG recordings that change over time. By eliminating unnecessary data, these gates enable LSTMs to hold onto significant information for extended periods of time. LSTM models are able to detect subtle shifts between normal and abnormal brain activity in seizure detection by efficiently learning from the dynamic changes in EEG signal patterns across time windows. LSTM networks provide a potent tool for real-time monitoring and early warning systems in clinical settings by improving the classification accuracy for differentiating seizure episodes from non-seizure intervals while maintaining temporal context.

F. Hybrid CNN and LSTM

The hybrid CNN + LSTM model is especially useful for evaluating EEG signals in epileptic seizure detection because it combines the temporal sequence learning capability of LSTM networks with the spatial feature extraction power of Convolutional Neural Networks (CNN).

High-level spatial features, like frequency patterns and localized signal variations, are first extracted from EEG signals using CNN layers in this architecture. After that, these characteristics are sent to the LSTM layers, which simulate temporal dependencies over time and record the changes in brain activity before and during a convulsion. Through this integration, the model is able to comprehend the local features of EEG waveforms as well as how they change over time, leading to more reliable and accurate classification. By taking into account both temporal and spatial complexities, the hybrid approach performs better than traditional models and is ideal for clinical neuroscience diagnostic and real-time monitoring applications.

IV. Implementation and Testing

Accuracy, precision, recall, and F1-score are the main evaluation metrics used in the Table I to compare the performance of CNN, LSTM, and a hybrid CNN + LSTM model. Reliable classification performance based on spatial features is demonstrated by the CNN model's 93.5% accuracy, 92.1% precision, 91.6% recall, and 91.8% F1-score. With 94.7% accuracy, 93.8% precision, 93.4% recall, and a 93.6% F1-score, the LSTM model performs marginally better, demonstrating its ability to capture temporal dependencies in sequential EEG data.

TABLE I. Classification Performance

Metric	CNN	LSTM	CNN + LSTM
Accuracy	93.5	94.7	96.8
Precision	92.1	93.8	96.2
Recall	91.6	93.4	95.7
F1-Score	91.8	93.6	95.9

With the best accuracy of 96.8%, precision of 96.2%, recall of 95.7%, and F1-score of 95.9%, the hybrid CNN + LSTM model surpasses both, proving that combining spatial and temporal features greatly improves the model's capacity to precisely identify and categorize epileptic seizures.

The model's classification performance for three classes—Rust disease, Healthy, and Downy mildew—is represented by the ROC curve in Fig.2, which has AUC values of 0.49, 0.51, and 0.51 for each class, respectively. These numbers are extremely near to 0.5, suggesting that the model's discriminatory ability is weak and that it only marginally outperforms random guessing. A robust classifier should ideally produce AUC values that are near 1.0, indicating that it can reliably distinguish between classes. The ROC plot's curves close to the diagonal line highlight the model's ineffectiveness at class separation. To improve accuracy and reliability in disease classification, this result emphasizes the need for better feature engineering, better model optimization, or more complex architectures like CNN, LSTM, or hybrid deep learning frameworks.

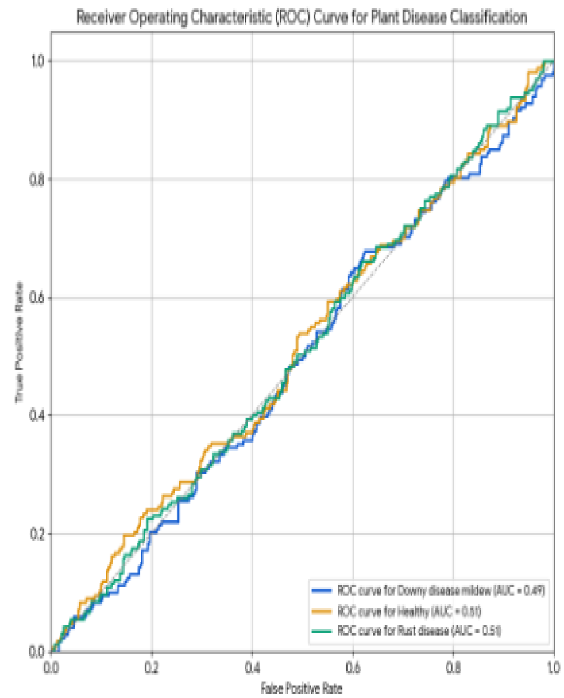

Fig. 2. ROC Curve for Plant Disease Classification

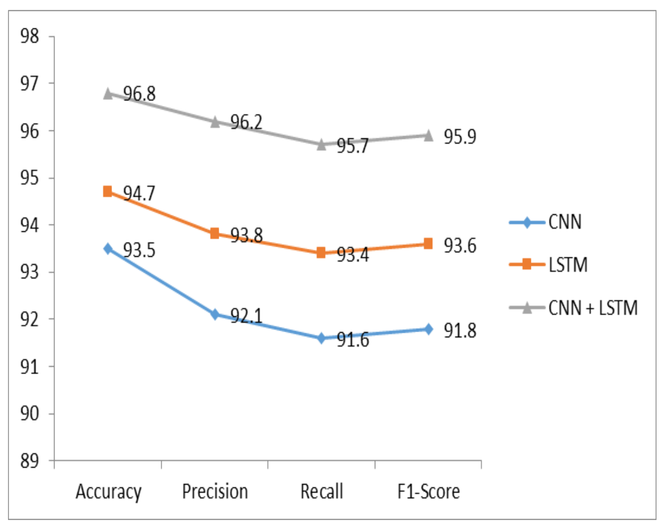

Fig. 3. Performance Comparison Chart

Accuracy, precision, recall, and F1-score are important metrics that are used to visually compare the performance of CNN, LSTM, and a hybrid CNN + LSTM model in Fig.3. With the highest scores, including an Accuracy of 96.8% and an F1-Score of 95.9%, the Hybrid CNN + LSTM architecture clearly outperforms its standalone counterparts on all assessed metrics. While the CNN-only model still performs well, it performs marginally worse than the LSTM model, which typically takes second place. This obvious difference demonstrates how convolutional layers for reliable feature extraction and LSTM layers for capturing temporal dependencies work in concert, making the hybrid approach the most efficient for the given task.

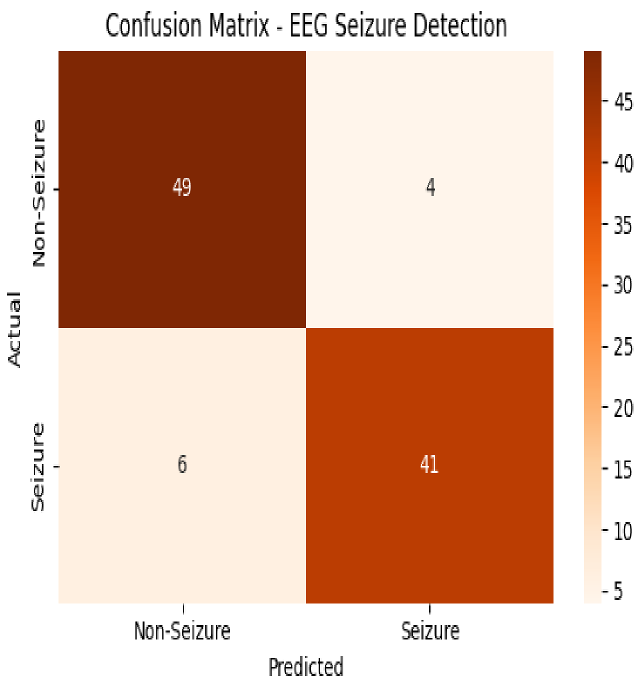

Fig. 4. Confusion Matrix- EEG Seizure Detection

The EEG seizure detection model's confusion matrix in Fig.4 demonstrates its excellent classification performance in both seizure and non-seizure categories. The model demonstrated a high degree of accuracy by correctly identifying 41 seizure cases and 49 non-seizure cases. There were very few misclassifications; only four non-seizure cases were mistakenly classified as seizures (false positives), and six seizure cases were overlooked (false negatives).

These findings demonstrate the model's resilience in correctly identifying seizures while preserving a low false alarm rate. In actual clinical settings, this type of performance is crucial because it guarantees prompt, accurate seizure detection with few misdiagnosis errors, which eventually supports improved patient monitoring and intervention.

V. CONCLUSION AND FUTURE WORK

The superior performance of the hybrid approach is evident from the comparative analysis of CNN, LSTM, and their hybrid CNN-LSTM model for epileptic seizure detection using EEG signals. The CNN-LSTM model performs better than the individual CNN and LSTM architectures in every important metric, with accuracy of 96.8%, precision of 96.2%, recall of 95.7%, and F1-score of 95.9%. This suggests that both spatial and temporal dependencies in EEG data can be efficiently captured by combining the spatial feature extraction capabilities of convolutional layers with the temporal sequence learning of LSTM. The performance and interpretability of classification could be further enhanced in the future by adding transformer-based models or attention mechanisms. Furthermore, generalizing the strategy for practical clinical implementation may be facilitated by extending the model to manage multi-channel EEG data from bigger and more varied datasets.

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